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Detecting forest disturbances in mid-latitude ecosystems using Sentinel-2 time series of functional traits anomalies

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***Abstract*—Forest are facing increasing stress from climate-induced droughts and extreme weather events, threatening ecosystem function and resilience. Timely detection of these disturbances is crucial for effective forest management. Remote sensing offers scalable monitoring of forest condition and dynamics across large areas with consistent spatial and temporal resolution. While most forest disturbance mapping approaches rely on indirect proxies like vegetation indices, this study explores the potential of Sentinel-2 derived plant traits, Canopy Chlorophyll Content (CCC) and Leaf Area Index (LAI), which offer more direct links to vegetation function and structure. Two documented disturbance types were analyzed: the 2022 summer drought and 2023 windthrow. A rolling regression slope analysis was applied to cumulative anomalies of functional traits to assess the timing, intensity, and persistence of the disturbances across ground-validated plots. Results show droughts generate isolated anomaly peaks during the event, with a stronger impact on leaf pigmentation (average reduction in CCC of -41.53% *vs.* -31.93% in LAI). In contrast, treefall produces a comparable reductions in both traits (exceeding 30.00%), indicating both functional and structural canopy loss. In the year following the event, a prolonged plateau of anomalies is observed, with partial mid-season attenuation linked to understory dynamics. These findings highlight the potential of the proposed approach to effectively support long-term forest monitoring and conservation strategies.**

***Index Terms*—Plant functional traits, anomaly detection, forest disturbance mapping, drought and treefall, multispectral satellite imagery**

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I. Introduction

Covering nearly one-third of the Earth's surface, forests deliver key ecosystem services such as climate regulation, biodiversity conservation, carbon sequestration and support for livelihoods through timber and food production [1]. However, they are increasingly exposed to a range of disturbances including droughts, storms, late frost, fires, and disease outbreaks, which can significantly alter their structure, function and long-term stability [2]. With the intensification of extreme climatic events, understanding their impacts on forest dynamics has become critical for developing adaptive management strategies. In this context, remote sensing technologies play a pivotal role by enabling continuous and large-scale monitoring of forest condition over time [3]. Over the years, multispectral sensors such as Landsat, MODIS and Sentinel-2 have been widely used to monitor forest structure, function, vitality and loss dynamics [4]. Nevertheless, several challenges persist in the detection and interpretation of forest disturbances. These include the lack of high-quality reference data for validation, the overlapping effects of multiple stressors in complex forest environments, and the limited transferability of models across different ecological zones and forest types [5]- [6]. Most approaches for mapping forest disturbances rely on time series of indirect proxies, employing temporal segmentation

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or multi-year deviations from expected phenological trends [5]- [7]- [8]. These methods often depend on subjective thresholds, require identifying undisturbed baseline periods, and frequently use a single annual observations or bi-temporal differencing. Furthermore, most existing studies rely on time series of vegetation indices or specific spectral bands [9]. Although widely used, they often suffer from saturation in dense canopies, sensitivity to atmospheric conditions, viewing geometry, and limited physiological interpretation. In contrast, biophysical canopy traits such as canopy chlorophyll content (CCC) and Leaf Area Index (LAI) provide direct indicators of changes in vegetation structure and function, as they are closely linked to photosynthetic activity, and overall plant health. This makes them particularly well-suited for characterizing stress-induced responses [10]. Building on this rationale, our analysis serves as an initial step towards the broader goals of the EO4Resilience project, funded by the European Space Agency (ESA) under the Sentinel Users Preparation (SUP) initiative. The project aims to advance development of innovative EO-based approaches for assessing forest resilience to climate-induced disturbances such as drought, with a particular emphasis on linking remote sensing observations to key functional canopy traits. Within this wider context, this study exploits Sentinel-2 time series of CCC and LAI traits to provide an operational case study on how biophysical canopy indicators can be used detect forest disturbance dynamics and evaluate post event recovery, based on documented 2022-2023 stress events. Specifically, the objectives are to i) detect and distinguish drought- and treefall-induced disturbances, and ii) assessing post-disturbance recovery patterns to evaluate forest resilience associated with different stress types.

II. Materials and Methods

A. Study area

The study was conducted in the Ticino Regional Park ($45.4286°$ N, $8.8342°$ E), one of the largest remaining natural mixed temperate forests in the Po valley, northern Italy. Covering more than 900 km^2, the park hosts fluvial ecosystems, riparian woodlands and rare remnants of the primary plain forest that once dominated the region [11]. The ecological diversity makes the area a key reservoir of diverse habitat and plant species [11] - [12]. The northern part is primarily composed of mesophylic mixed oak-hornbeam association, with *Quercus robur* and *Carpinus betulus* as dominant species, sweet chestnut (*Castanea sativa*) and conifer species (*Pinus silvestris* and *P. rigida*). The southern part is commonly populated by poplar (*Populus spp.*) and black alder (*Alnus glutinosa*). In recent years, northern Italy was identified as a hotspot of one of the most severe droughts across southern Europe, which occurred in 2022 [13]. Hence, the Ticino Regional Park experienced elevated forest stress and mortality, which caused widespread canopy degradation and early leaf senescence. Drought impacts were found uneven across forest types [10]: pine and black cherry showed the strongest and most prolonged stress responses, while xerophilic oak and mixed deciduous forests also indicated moderate to severe stress. In contrast, hygrophilic species like black alder, chestnut, meso-hygrophylic oak associated with white hornbeam showed minimal changes, suggesting higher resilience. In addition, during the summer 2023 severe convective storms, notably on July 23-27 and August 26, hit the region leading to localized windthrow events. According to the regional environmental protection agency ARPA Lombardia, extreme weather events were characterized by wind gusts exceeding 100 km/h, contributing to blowdowns and treefalls throughout the park [14].

B. Disturbance reference data

Reference data, used to validate the method for forest disturbance assessment, were collected during a field campaign in summer 2022 from homogeneous forest type plots (30x30 m), following protocols described by [10] for plant dominance and species composition. Starting from the location of reference data, 2022 drought-affected areas were identified according to observed high anomalies in validated vegetation traits in relation to meteorological data [10], while zones impacted by 2023 treefall were determined through direct on-site inspections carried out by the park's ecologists and botanical experts, who recorded photographs and visible evidence of windthrow damage.

C. Standardized anomalies of functional traits

Following the methodology presented in [10], the detection of forest traits anomalies was carried out from a total of 230 cloud-free Sentinel-2 multispectral images collected between 2017 and 2024. Functional vegetation traits of CCC and LAI were extracted using the biophysical processor of the S2 toolbox [15] integrated within the Sentinel Application Platform (SNAP) software. This processor employs a hybrid retrieval algorithm combining artificial neural networks (ANN) and radiative transfer models (PROSPECT for leaf-level and SAIL for canopy simulations) to estimate trait values from reflectance data [15] - [16]. The processor has demonstrated strong agreement with reference data in the study area with CCC and LAI estimations achieving nRMSE values of 13.56% and 11.49%, respectively [10]. Raw time series of CCC and LAI were extracted in 63 sampling plots from forest traits maps, and smoothed with a Savitzky-Golay filter [17] to reduce noise and low-quality observations. Then, a winter baseline adjustment was applied by replacing values from Day of Year (DOY) 335 to 56 with the 5th percentile of the entire smoothed time series, to minimize noise from snow, cloud and standardize seasonal dormancy. Finally, 5-days interpolation was performed using a spline function to ensure continuity and produce regularly spaced time series. Anomalies in forest traits were then identified using the Standardized Anomaly (SA), calculated as the deviation of each 5-day observation from the corresponding multi-year (2017–2024) average, normalized by its standard deviation. This approach allowed the generation of consistent 5-day time series for both CCC and LAI for each plot.

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D. Forest disturbance detection

To assess forest disturbances associated with drought and treefall events, the Cumulative SA (CSA) was computed for each sampling plot over the 2017-2024 period as the sum of SA values up to time (1). Unlike instantaneous anomalies, CSA captures the persistence and intensity of functional trait deviations, highlighting periods of vegetation ecological stress.

$$\mathrm{CSA}_t = \sum_{i=1}^{t} \mathrm{SA}_i \tag{1}$$

Furthermore, to characterize rapid deviations and potential recovery phases, the rate of change of CSA was assessed using a rolling regression slope. Linear regression of each CSA time series was applied within a fixed-size moving window of width *(w)*, generating a sequence of local slopes that quantify the rate and direction of change over time. Negative slopes indicate periods of increasing stress accumulation, while positive slopes suggest recovery or stabilization. The time window size was empirically set to 10 weeks, based on ecological considerations and sensitivity tests. This duration is long enough to capture the typical progression of drought induced stress, yet short enough to detect abrupt disturbances such as treefall. Testing alternative window lengths (i.e. 5, 10, 15 and 20 weeks) confirmed that 10 weeks provided a good balance between temporal sensitivity and noise reduction. The slope at each central time point *t* within the moving window is calculated as shown in (2),

$$\mathrm{RollingSlope}_t = \frac{\sum_{i=0}^{w-1}(t_i - \bar{t})(CSA_i - C\bar{S}A)}{\sum_{i=0}^{w-1}(t_i - \bar{t})^2} \tag{2}$$

where t_i is the time index, CSA_i is the CSA at time t_i, while $\bar{t}$ and $C\bar{S}A$ are the mean values in the temporal window. Starting from the rolling regression slope time series, segments with negative slopes were selected, indicating a declining trend in vegetation condition. Within each segment, the local minimum slope falling below a defined threshold was selected as the point of peak disturbance intensity. The corresponding date was then recorded as the moment of maximum stress experienced by the vegetation.

III. Results

This study focuses on confirmed disturbance events, specifically the 2022 drought and 2023 treefall, to evaluate the method's ability to identify such events based on their effects on forest condition. Fig. 1a shows selected plots within a central area of the park, all of which were affected by the 2022 drought. Among these, some experienced treefall in 2023 (FOR07, FOR16, FOR19, FOR21, FOR52, FOR53, FOR54, FOR57, FOR60, FOR74). These plots also represent a range of dominant vegetation species, including meso-hygrophilic oak or associated with white hornbeam, mixed deciduous, open xerophilic oak and hygrophilic black alder. As shown in Fig. 1b, the CCC reconstructed time series for plot FOR71 (drought-affected) exhibits clear deviations from the multi-year average (2017-2024) during late summer of 2022 and the following spring, indicating marked disturbance characterized by a reduction in the forest trait value. Similarly, Fig. 1c presents LAI time series for a plot affected by late summer treefall disturbance in 2023 (FOR19), highlighting the impact on forest canopy structure and a consequent decline in the respective trait estimation. Notably, in the case of treefall (Fig. 1c), the trait anomaly persists into the following year (2024), as the physical absence of the affected tree prevents any structural recovery at that location, resulting in a sustained deviation from the baseline value.

As shown in Fig. 2, rolling regression slope of CCC for drought-affected plot (FOR71) reaches its minimum value with intensity around -1.99 during summer 2022, coinciding with the peak of the drought period. The slope subsequently increases, turning positive in autumn, suggesting possible vegetation recovery. A second local minimum with intensity around -1.95 is observed in May 2023, although its attribution remains uncertain and could potentially be linked to a late frost event. A third peak in June 2024 might be associated to a secondary, less severe drought episode compared to 2022. In contrast, rolling regression slope of LAI calculated for treefall affected plot (FOR19) in the same figure (Fig. 2) reveals a pronounced local minimum nearly aligned with the second recorded storm date in summer 2023, with an intensity peak of -1.57, reflecting immediate structural loss due to canopy collapse. The persistence of negative anomalies from the multi-year (2017-2024) average LAI time series is observed in the following 2024 vegetative period forming a broad plateau with intensity around -2.30. A slight mid-season attenuation of the anomaly is observed, which may be attributed to the temporary contribution of understory species, such as herbaceous plants and shrubs, that green up during summer and contribute to the optical signal. As these species enter senescence in late summer and early autumn, the anomaly deepens again, reinforcing the overall negative peak.

The rolling slope analysis for CCC and LAI forest traits expanded to the 2020-2024 time range and across all sites is reported in Fig. 3. A threshold of -1.0 in peak intensity was applied to identify meaningful disturbance events. In plots affected by 2022 drought (blue dots in Fig. 3), isolated negative peaks in both CCC and LAI are visible (Fig. 3a and Fig. 3b), reflecting sharp and temporally localized reduction in forest function, more accentuated for CCC than LAI. These drought-related effects tend to occur at variable times across plots, indicating heterogeneous sensitivity and response dynamics among sites and species. After 2022, other peaks are observed in summer 2023 and 2024, suggesting other drought-related effects. In contrast, treefall-affected plots (red dots in Fig. 3) display a consistent temporal clustering of disturbance around summer 2023 storm events, both for CCC and LAI, confirming the method to point timing of canopy loss. Furthermore, many of these points exhibit a prolonged negative anomaly throughout the 2024 growing season, forming a characteristic plateau in both CCC and LAI (Fig. 3a and Fig. 3b). Within

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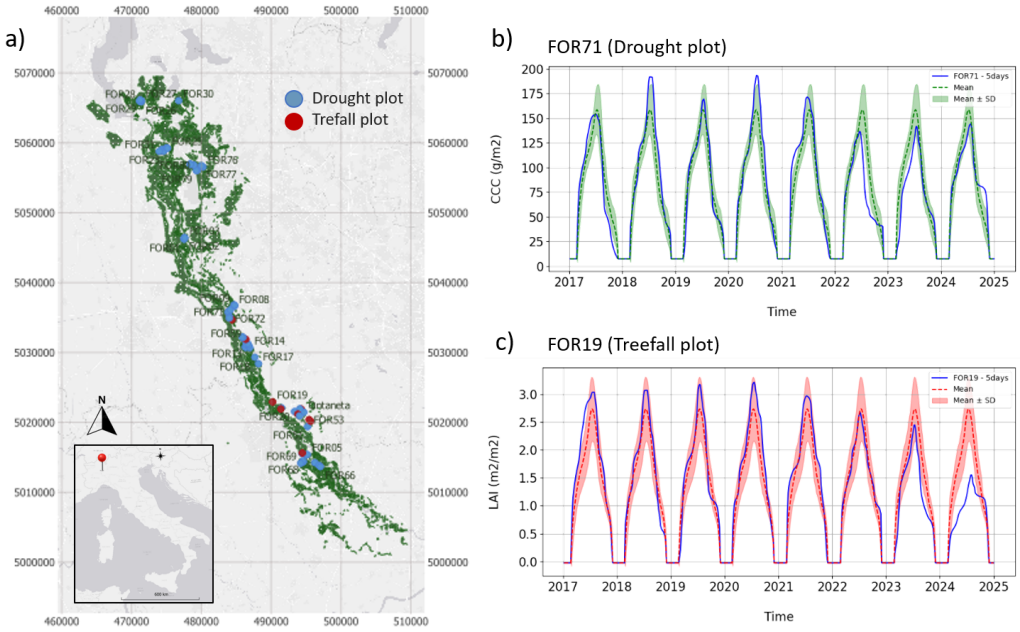


Fig. 1. a) Map of the study area location within the central portion of the Ticino Regional Park, showing sampling plots predominandly affected by 2022 drought (blue) and 2023 treefall (red). Alongside blue lines for b) 5-day time series of canopy chlorophyll content (CCC) for FOR71 representative of the 2022 drought and c) 5-days time series of leaf area index (LAI) for FOR19 impacted by treefall events in 2023, compared to the corresponding CCC and LAI multi-year average (2017-2024) and standard deviation time series (green and red lines, respectively). Map coordinate reference system: WGS84 UTM zone 32N.

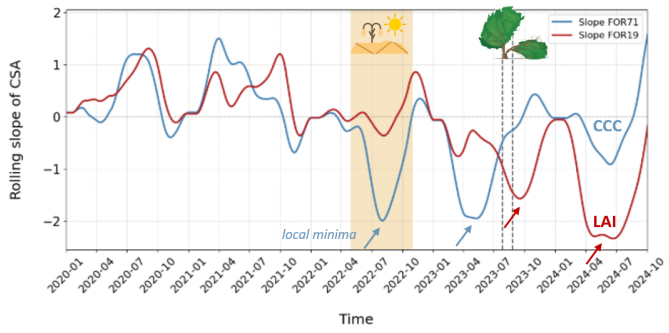


Fig. 2. Example of rolling regression slopes of CSA (Cumulative sum of Standardized Anomalies) for CCC in drought-affected plot (FOR71) and for LAI in treefall-affected plot (FOR19). The 2022 drought period and 2023 summer storm events are highlighted.

this plateau, a slight mid-season attenuation is observed, likely reflecting gap dynamics influenced by understory vegetation development. This temporal signature of peaks differentiates treefall disturbances from drought, which generally produces mainly singular anomalies without prolonged responses.

To identify potential distinguishing patterns in CCC and LAI dynamics associated with different disturbance types, Table I reports the average percentage reduction in CCC and LAI relative to the multi-year mean values, calculated at the time of the disturbance event or at nearby dates in subsequent

TABLE I
AVERAGE AND STANDARD DEVIATION OF THE PERCENTAGE REDUCTION IN CCC AND LAI RELATIVE TO THE MULTI-YEAR AVERAGE CALCULATED IN THE DISTURBANCE DATES OR IN THE SUBSEQUENT YEARS.

Disturbance	Year	CCC Reduction (%)	LAI Reduction (%)
Treefall	2023 (event year)	-32.56 ± 10.80	-33.76 ± 8.04
Treefall	2024 (1 year later)	-14.07 ± 11.82	-16.82 ± 5.83
Drought	2022 (event year)	-41.53 ± 13.32	-31.93 ± 11.54
Drought	2023 (1 year later)	-3.23 ± 11.98	-3.01 ± 9.60
Drought	2024 (2 years later)	-5.59 ± 13.93	-4.39 ± 10.46

years across the selected plots. Results show that for treefall in the event year (2023) the average CCC reduction was -32.56%, similar to LAI. This reflects how a decrease in canopy coverage simultaneously affects chlorophyll content and leaf area. One year later (2024), reduction in both parameters were considerably smaller (around -15.00%), indicating partial recovery, likely due to understory regrowth. The persistence of a modest reduction suggests that full canopy recovery may require multiple years. In contrast, drought event in 2022 caused a larger immediate average CCC reduction in the event year (2022), higher than LAI reduction (-41.53% vs. -31.93%). This pattern is consistent with physiological stress responses, where color loss precedes actual leaf drops. In the following two years (2023 and 2024), both CCC and LAI showed substantial recovery, although a mild negative values in 2024 may suggest either residual drought effects. Overall,

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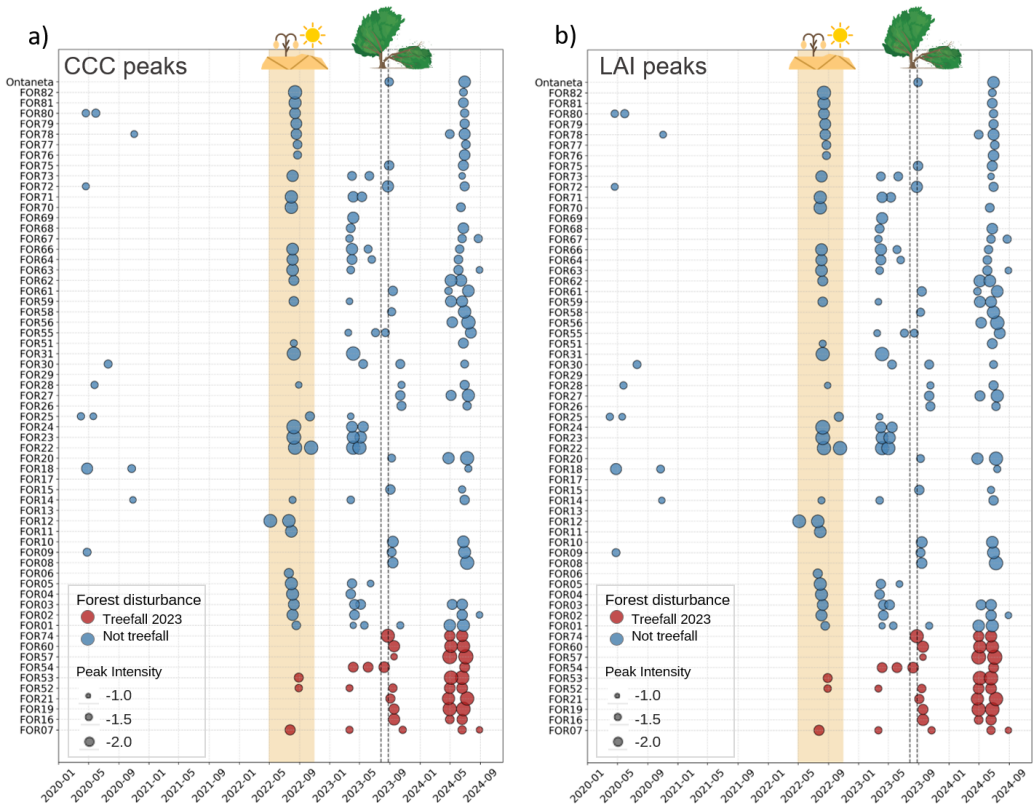


Fig. 3. Timing and intensity of vegetation disturbance events based on rolling slope analysis for a) CCC and b) LAI, across monitored forest plots affected by drought (blue dots) or treefall (red dots). Each point represent a local minimum in the slope time series, with size scaled by peak intensity (only values $<$ -1.0 shown).

the results highlight clear differences in recovery dynamics between the two types of disturbances: treefall causes abrupt proportional reductions in both CCC and LAI that do not show clear recovery in the following year, while drought impacts CCC more strongly than LAI in the event year, with relatively rapid recovery in the following years.

IV. Conclusion

These preliminary results show the potential of Sentinel-2 derived CCC and LAI time series to detect and differentiate forest disturbances from drought and treefall in temperate ecosystems. Rolling slope analysis of cumulative anomalies seems to identify with good accuracy the timing of disturbance peak, aligning with documented drought and storm events. Drought were mostly characterized by isolated negative peaks, particularly in CCC, reflecting short-term physiological stress. In contrast, treefall led to a initial peak at the time of the event, followed by a prolonged plateau of anomalies, with mid-season attenuation due to understory dynamics. Percentage reductions confirmed both CCC and LAI experienced significant decline, with CCC more affected by drought, due to pigment degradation, and both traits equally impacted by treefall due to structural canopy damage. Recovery patterns showed a gradual return to baseline after drought, whereas the effects of treefall persisted for a longer period. These results present a first step towards the detection of forest stress from biophysical traits across multiple disturbance and forest types. Nevertheless, some limitations should be considered: the ANNs implemented in the Sentinel-2 Toolbox are trained on simulated data, so their accuracy depends on how well simulations reflect real leaves, canopy and soil conditions. In addition, retrievals are sensitive to input quality (i.e. Sentinel-2 reflectance), meaning that presence of clouds, atmospheric effects, or sensor calibration errors may propagate into the final outputs. Moreover, cloud cover and variable atmospheric conditions can reduce the availability of observations, potentially introducing gaps or noise that may affect time series reconstruction and anomaly estimates. Future developments could explore the generalization of the disturbance detection framework across different forest types and geographic regions. The CSA and rolling slope approach is conceptually flexible, as it only relies on temporal dynamics of biophysical traits, making it potentially transferable to forest with contrasting stuctures and phenologies. Nevertheless, adapting the methodology to diverse ecosystems would require validation against local ground measurement and consideration about region-specific disturbances (e.g. fires, windthrow, cyclones). Another promising avenue is the extension to a broader set of

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functional traits, beyond those investigated here, by exploiting existing and upcoming hyperspectral satellite missions (e.g. PRISMA, EnMAP, future CHIME) that provide richer spectral information for canopy chemistry and physiology. Future work could also extend the approach by integrating complementary data streams. For instance, existent and upcoming radar observations like Sentinel-1 and future ROSE-L may provide robust information on canopy structure and soil moisture under frequent cloud cover, while LiDAR data could capture additional dimensions of forest stress and recovery. Combining Sentinel-2 with these sources would strengthen the resilience and scalability of the method for global forest monitoring.

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